

1. a.

$$\begin{aligned}\det A &= a_{31}C_{31} + a_{35}C_{35} \\ &= 3C_{31} + (-1)C_{35}\end{aligned}$$

$$\begin{aligned}M_{31} &= -4[(-2)(-6) - 4(-21) + 6(-39)] \\ &\quad + 8[4(-6) - 4(-12) + 6(-18)] \\ &\quad + (-3)[4(-39) - (-2)(-18) + 4(15)] \\ &= -4(-138) + 8(-84) + (-3)(-132) = 276\end{aligned}$$

$$C_{31} = (-1)^{3+1}(276) = 276$$

$$\begin{aligned}M_{35} &= 1[3(-24) - (-1)(-12) + 9(8)] \\ &= 1(-12) = -12\end{aligned}$$

$$C_{35} = (-1)^{3+5}(-12) = -12$$

$$\begin{aligned}\det A &= 3(276) + (-1)(-12) \\ &= 828 + 12\end{aligned}$$

$$\boxed{\det A = 840}$$

b.

$$\begin{aligned}M_{11} &= (-1)[3(-24) - (-1)(-12) + 9(8)] \\ &= (-1)(-12) = 12\end{aligned}$$

$$\begin{aligned}M_{12} &= -3[(-1)(-9) - 9(-20) + 5(-24)] \\ &= -3(69) = -207\end{aligned}$$

$$\begin{aligned}M_{13} &= -3[3(-9) - 9(-13) + 5(-12)] \\ &= -3(30) = -90\end{aligned}$$

$$\begin{aligned}M_{14} &= -3[3(-20) - (-1)(-13) + 5(8)] \\ &= -3(-33) = 99\end{aligned}$$

$$\begin{aligned}M_{15} &= -3[3(-24) - (-1)(-12) + 9(8)] \\ &= -3(-12) = 36\end{aligned}$$

$$\begin{aligned}M_{21} &= (-1)[4(-24) - (-2)(-12) + 4(8)] \\ &= (-1)(-88) = 88\end{aligned}$$

$$\begin{aligned}M_{22} &= -3[(-2)(-9) - 4(-20) + 6(-24)] \\ &\quad + (-1)[1(-24) - (-2)(0) + 4(0)] \\ &= -3(-46) + (-1)(-24) = 162\end{aligned}$$

$$\begin{aligned}M_{23} &= -3[4(-9) - 4(-13) + 6(-12)] \\ &\quad + (-1)[1(-12) - 4(0) + 4(0)] \\ &= -3(-56) + (-1)(-12) = 180\end{aligned}$$

$$\begin{aligned}M_{24} &= -3[4(-20) - (-2)(-13) + 6(8)] \\ &\quad + (-1)[1(8) - 4(0) + (-2)(0)] \\ &= -3(-58) + (-1)(8) = 166\end{aligned}$$

$$\begin{aligned}M_{25} &= -3[4(-24) - (-2)(-12) + 4(8)] \\ &= -3(-88) = 264\end{aligned}$$

$$\begin{aligned}M_{31} &= -4[(-2)(-6) - 4(-21) + 6(-39)] \\ &\quad + 8[4(-6) - 4(-12) + 6(-18)] \\ &\quad + (-3)[4(-39) - (-2)(-18) + 4(15)] \\ &= -4(-138) + 8(-84) + (-3)(-132) = 276\end{aligned}$$

$$\begin{aligned}M_{32} &= 1[(-1)(-9) - 9(-20) + 5(-24)] \\ &= 1(69) = 69\end{aligned}$$

$$\begin{aligned}M_{33} &= 1[3(-9) - 9(-13) + 5(-12)] \\ &= 1(30) = 30\end{aligned}$$

$$\begin{aligned}M_{34} &= 1[3(-20) - (-1)(-13) + 5(8)] \\ &= 1(-33) = -33\end{aligned}$$

$$\begin{aligned}M_{35} &= 1[3(-24) - (-1)(-12) + 9(8)] \\ &= 1(-12) = -12\end{aligned}$$

$$\begin{aligned}M_{41} &= -(-1)[4(-72) - (-2)(-36) + 4(28)] \\ &= -(-1)(-248) = -248\end{aligned}$$

$$\begin{aligned}M_{42} &= 3[(-2)(-27) - 4(-37) + 6(-72)] \\ &\quad - (-1)[1(-72) - (-2)(0) + 4(0)] \\ &= 3(-230) - (-1)(-72) = -762\end{aligned}$$

$$\begin{aligned}M_{43} &= 3[4(-27) - 4(-29) + 6(-36)] \\ &\quad - (-1)[1(-36) - 4(0) + 4(0)] \\ &= 3(-208) - (-1)(-36) = -660\end{aligned}$$

$$\begin{aligned}M_{44} &= 3[4(-37) - (-2)(-29) + 6(28)] \\ &\quad - (-1)[1(28) - 4(0) + (-2)(0)] \\ &= 3(-38) - (-1)(28) = -86\end{aligned}$$

$$\begin{aligned}M_{45} &= 3[4(-72) - (-2)(-36) + 4(28)] \\ &= 3(-248) = -744\end{aligned}$$

$$\begin{aligned}M_{51} &= -(-1)[4(-39) - (-2)(-18) + 4(15)] \\ &= -(-1)(-132) = -132\end{aligned}$$

$$\begin{aligned}M_{52} &= 3[(-2)(-6) - 4(-21) + 6(-39)] \\ &\quad - (-1)[1(-39) - (-2)(0) + 4(0)] \\ &= 3(-138) - (-1)(-39) = -453\end{aligned}$$

$$\begin{aligned}
M_{53} &= 3[4(-6) - 4(-12) + 6(-18)] \\
&\quad - (-1)[1(-18) - 4(0) + 4(0)] \\
&= 3(-84) - (-1)(-18) = -270
\end{aligned}$$

$$\begin{aligned}
M_{54} &= 3[4(-21) - (-2)(-12) + 6(15)] \\
&\quad - (-1)[1(15) - 4(0) + (-2)(0)] \\
&= 3(-18) - (-1)(15) = -39
\end{aligned}$$

$$\begin{aligned}
M_{55} &= 3[4(-39) - (-2)(-18) + 4(15)] \\
&= 3(-132) = -396
\end{aligned}$$

$$M = \begin{bmatrix} 12 & -207 & -90 & 99 & 36 \\ 88 & 162 & 180 & 166 & 264 \\ 276 & 69 & 30 & -33 & -12 \\ -248 & -762 & -660 & -86 & -744 \\ -132 & -453 & -270 & -39 & -396 \end{bmatrix}$$

c.

$$C_{ij} = (-1)^{i+j} M_{ij}$$

$$C = \begin{bmatrix} 12 & 207 & -90 & -99 & 36 \\ -88 & 162 & -180 & 166 & -264 \\ 276 & -69 & 30 & 33 & -12 \\ 248 & -762 & 660 & -86 & 744 \\ -132 & 453 & -270 & 39 & -396 \end{bmatrix}$$

d.

$$\text{adj } A = C^T = \begin{bmatrix} 12 & -88 & 276 & 248 & -132 \\ 207 & 162 & -69 & -762 & 453 \\ -90 & -180 & 30 & 660 & -270 \\ -99 & 166 & 33 & -86 & 39 \\ 36 & -264 & -12 & 744 & -396 \end{bmatrix}$$

e.

$$\begin{aligned}
R_3 &\rightarrow R_3 - 3R_1 \\
&= [0 \ -12 \ 6 \ -12 \ -19 \ | \ -3 \ 0 \ 1 \ 0 \ 0]
\end{aligned}$$

$$\begin{aligned}
R_2 &\rightarrow \frac{1}{3}R_2 \\
&= [0 \ 1 \ -\frac{1}{3} \ 3 \ \frac{5}{3} \ | \ 0 \ \frac{1}{3} \ 0 \ 0 \ 0]
\end{aligned}$$

$$\begin{aligned}
R_1 &\rightarrow R_1 - 4R_2 \\
&= [1 \ 0 \ -\frac{2}{3} \ -8 \ -\frac{2}{3} \ | \ 1 \ -\frac{4}{3} \ 0 \ 0 \ 0]
\end{aligned}$$

$$\begin{aligned}
R_3 &\rightarrow R_3 + 12R_2 \\
&= [0 \ 0 \ 2 \ 24 \ 1 \ | \ -3 \ 4 \ 1 \ 0 \ 0]
\end{aligned}$$

$$\begin{aligned}
R_4 &\rightarrow R_4 - 3R_2 \\
&= [0 \ 0 \ 5 \ -6 \ -4 \ | \ 0 \ -1 \ 0 \ 1 \ 0]
\end{aligned}$$

$$\begin{aligned}
R_5 &\rightarrow R_5 - 4R_2 \\
&= [0 \ 0 \ \frac{28}{3} \ -12 \ -\frac{29}{3} \ | \ 0 \ -\frac{4}{3} \ 0 \ 0 \ 1]
\end{aligned}$$

$$\begin{aligned}
R_3 &\rightarrow \frac{1}{2}R_3 \\
&= [0 \ 0 \ 1 \ 12 \ \frac{1}{2} \ | \ -\frac{3}{2} \ 2 \ \frac{1}{2} \ 0 \ 0]
\end{aligned}$$

$$\begin{aligned}
R_1 &\rightarrow R_1 + \frac{2}{3}R_3 \\
&= [1 \ 0 \ 0 \ 0 \ -\frac{1}{3} \ | \ 0 \ 0 \ \frac{1}{3} \ 0 \ 0]
\end{aligned}$$

$$\begin{aligned}
R_2 &\rightarrow R_2 + \frac{1}{3}R_3 \\
&= [0 \ 1 \ 0 \ 7 \ \frac{11}{6} \ | \ -\frac{1}{2} \ 1 \ \frac{1}{6} \ 0 \ 0]
\end{aligned}$$

$$\begin{aligned}
R_4 &\rightarrow R_4 - 5R_3 \\
&= [0 \ 0 \ 0 \ -66 \ -\frac{13}{2} \ | \ \frac{15}{2} \ -11 \ -\frac{5}{2} \ 1 \ 0]
\end{aligned}$$

$$\begin{aligned}
R_5 &\rightarrow R_5 - \frac{28}{3}R_3 \\
&= [0 \ 0 \ 0 \ -124 \ -\frac{43}{3} \ | \ 14 \ -20 \ -\frac{14}{3} \ 0 \ 1]
\end{aligned}$$

$$\begin{aligned}
R_4 &\rightarrow -\frac{1}{66}R_4 \\
&= [0 \ 0 \ 0 \ 1 \ \frac{13}{132} \ | \ -\frac{5}{44} \ \frac{1}{6} \ \frac{5}{132} \ -\frac{1}{66} \ 0]
\end{aligned}$$

$$\begin{aligned}
R_2 &\rightarrow R_2 - 7R_4 \\
&= [0 \ 1 \ 0 \ 0 \ \frac{151}{132} \ | \ \frac{13}{44} \ -\frac{1}{6} \ -\frac{13}{132} \ \frac{7}{66} \ 0]
\end{aligned}$$

$$\begin{aligned}
R_3 &\rightarrow R_3 - 12R_4 \\
&= [0 \ 0 \ 1 \ 0 \ -\frac{15}{22} \ | \ -\frac{3}{22} \ 0 \ \frac{1}{22} \ \frac{2}{11} \ 0]
\end{aligned}$$

$$\begin{aligned}
R_5 &\rightarrow R_5 + 124R_4 \\
&= [0 \ 0 \ 0 \ 0 \ -\frac{70}{33} \ | \ -\frac{1}{11} \ \frac{2}{3} \ \frac{1}{33} \ -\frac{62}{33} \ 1]
\end{aligned}$$

$$\begin{aligned}
R_5 &\rightarrow -\frac{33}{70}R_5 \\
&= [0 \ 0 \ 0 \ 0 \ 1 \ | \ \frac{3}{70} \ -\frac{11}{35} \ -\frac{1}{70} \ \frac{31}{35} \ -\frac{33}{70}]
\end{aligned}$$

$$\begin{aligned}
R_1 &\rightarrow R_1 + \frac{1}{3}R_5 \\
&= [1 \ 0 \ 0 \ 0 \ 0 \ | \ \frac{1}{70} \ -\frac{11}{105} \ \frac{23}{70} \ \frac{31}{105} \ -\frac{11}{70}]
\end{aligned}$$

$$\begin{aligned}
R_2 &\rightarrow R_2 - \frac{151}{132}R_5 \\
&= [0 \ 1 \ 0 \ 0 \ 0 \ | \ \frac{69}{280} \ \frac{27}{140} \ -\frac{23}{280} \ -\frac{127}{140} \ \frac{151}{280}]
\end{aligned}$$

$$\begin{aligned}
R_3 &\rightarrow R_3 + \frac{15}{22}R_5 \\
&= [0 \ 0 \ 1 \ 0 \ 0 \ | \ -\frac{3}{28} \ -\frac{3}{14} \ \frac{1}{28} \ \frac{11}{14} \ -\frac{9}{28}]
\end{aligned}$$

$$\begin{aligned}
R_4 &\rightarrow R_4 - \frac{13}{132}R_5 \\
&= [0 \ 0 \ 0 \ 1 \ 0 \ | \ -\frac{33}{280} \ \frac{83}{420} \ \frac{11}{280} \ -\frac{43}{420} \ \frac{13}{280}]
\end{aligned}$$

$$A^{-1} = \begin{bmatrix} \frac{1}{70} & -\frac{11}{105} & \frac{23}{70} & \frac{31}{105} & -\frac{11}{70} \\ \frac{69}{280} & \frac{27}{140} & -\frac{23}{280} & -\frac{127}{140} & \frac{151}{280} \\ -\frac{3}{28} & -\frac{3}{14} & \frac{1}{28} & \frac{11}{14} & -\frac{9}{28} \\ \frac{33}{280} & \frac{83}{420} & \frac{11}{280} & -\frac{43}{420} & \frac{13}{280} \\ \frac{3}{70} & -\frac{11}{35} & -\frac{1}{70} & \frac{31}{35} & -\frac{33}{70} \end{bmatrix}$$

2. a.

$$\begin{aligned}
R_2 &\rightarrow R_2 - 2R_1 \\
&= [(2 - 2(1)) \ (4 - 2(3)) \ (-1 - 2(4))] \\
&= [0 \ -2 \ -9]
\end{aligned}$$

$$\begin{aligned}
R_3 &\rightarrow R_3 - R_1 \\
&= [(1 - 1) \ (3 - 3) \ (-6 - 4)] \\
&= [0 \ 0 \ -10]
\end{aligned}$$

$$\text{REF} = \begin{bmatrix} 1 & 3 & 4 \\ 0 & -2 & -9 \\ 0 & 0 & -10 \end{bmatrix}$$

$$\det B = (1)(-2)(-10)$$

$$\boxed{\det B = 20}$$

b.

$$\det B = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$$

$$M_{11} = (4)(-6) - (-1)(3) = -21$$

$$M_{12} = (2)(-6) - (-1)(1) = -11$$

$$M_{13} = (2)(3) - (4)(1) = 2$$

$$\begin{aligned}
\det B &= 1(-21) - 3(-11) + 4(2) \\
&= -21 + 33 + 8
\end{aligned}$$

$$\boxed{\det B = 20}$$

c.

$$M_{11} = (4)(-6) - (-1)(3) = -21$$

$$M_{12} = (2)(-6) - (-1)(1) = -11$$

$$M_{13} = (2)(3) - (4)(1) = 2$$

$$M_{21} = (3)(-6) - (4)(3) = -30$$

$$M_{22} = (1)(-6) - (4)(1) = -10$$

$$M_{23} = (1)(3) - (3)(1) = 0$$

$$M_{31} = (3)(-1) - (4)(4) = -19$$

$$M_{32} = (1)(-1) - (4)(2) = -9$$

$$M_{33} = (1)(4) - (3)(2) = -2$$

$$\begin{aligned}
C &= \begin{bmatrix} +(-21) & -(-11) & +(2) \\ -(-30) & +(-10) & -(0) \\ +(-19) & -(-9) & +(-2) \end{bmatrix} \\
&= \begin{bmatrix} -21 & 11 & 2 \\ 30 & -10 & 0 \\ -19 & 9 & -2 \end{bmatrix}
\end{aligned}$$

$$\text{adj } B = C^T = \begin{bmatrix} -21 & 30 & -19 \\ 11 & -10 & 9 \\ 2 & 0 & -2 \end{bmatrix}$$

d.

$$\begin{aligned}
R_2 &\rightarrow R_2 - 2R_1 \\
&= [0 \ -2 \ -9 \mid -2 \ 1 \ 0]
\end{aligned}$$

$$\begin{aligned}
R_3 &\rightarrow R_3 - R_1 \\
&= [0 \ 0 \ -10 \mid -1 \ 0 \ 1]
\end{aligned}$$

$$\begin{aligned}
R_2 &\rightarrow -\frac{1}{2}R_2 \\
&= [0 \ 1 \ \frac{9}{2} \mid 1 \ -\frac{1}{2} \ 0]
\end{aligned}$$

$$\begin{aligned}
R_1 &\rightarrow R_1 - 3R_2 \\
&= [1 \ 0 \ -\frac{19}{2} \mid -2 \ \frac{3}{2} \ 0]
\end{aligned}$$

$$\begin{aligned}
R_3 &\rightarrow -\frac{1}{10}R_3 \\
&= [0 \ 0 \ 1 \mid \frac{1}{10} \ 0 \ -\frac{1}{10}]
\end{aligned}$$

$$\begin{aligned}
R_1 &\rightarrow R_1 + \frac{19}{2}R_3 \\
&= [1 \ 0 \ 0 \mid -\frac{21}{20} \ \frac{3}{2} \ -\frac{19}{20}]
\end{aligned}$$

$$\begin{aligned}
R_2 &\rightarrow R_2 - \frac{9}{2}R_3 \\
&= [0 \ 1 \ 0 \mid \frac{11}{20} \ -\frac{1}{2} \ \frac{9}{20}]
\end{aligned}$$

$$B^{-1} = \begin{bmatrix} -\frac{21}{20} & \frac{3}{2} & -\frac{19}{20} \\ \frac{11}{20} & -\frac{1}{2} & \frac{9}{20} \\ \frac{1}{10} & 0 & -\frac{1}{10} \end{bmatrix}$$

Verification:

$$\begin{aligned}
B^{-1} &= \frac{1}{\det B} \text{adj } B = \frac{1}{20} \begin{bmatrix} -21 & 30 & -19 \\ 11 & -10 & 9 \\ 2 & 0 & -2 \end{bmatrix} \\
&= \begin{bmatrix} -\frac{21}{20} & \frac{3}{2} & -\frac{19}{20} \\ \frac{11}{20} & -\frac{1}{2} & \frac{9}{20} \\ \frac{1}{10} & 0 & -\frac{1}{10} \end{bmatrix} \checkmark
\end{aligned}$$